

IMPROVED COOKSTOVES PROJECT FOR MALAWI AND CROSS-BORDER REGIONS OF MOZAMBIQUE – CPA – MAL – 005

Document Prepared By Ang Kong Nian

C-Quest Capital Malaysia Limited

Project Title	Improved Cookstoves Project for Malawi and cross-border regions of Mozambique – CPA – MAL - 005
Version	03
Date of Issue	27-November-2017
Monitoring Period	05-September-2016 to 31-July-2017
Prepared By	Mr. Ang Kong Nian, C-Quest Capital Malaysia Limited
Contact	Address: C-Quest Capital Malaysia Limited, Brumby Centre, Lot 42, Jalan Muhibbah, 87000 Labuan F. T., Malaysia, Telephone: +6 087 593828 Email: akongnian@cquestcapital.com Website: www.cquestcapital.com

Table of Contents

	Pg No.
1 PROJECT DETAILS	3
2 APPLICATION OF METHODOLOGY	9
3 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS	10
4 MONITORING	10
5 SAFEGUARDS	11
6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS	11

1 PROJECT DETAILS

1.1 Summary Description of the Project and its Implementation Status

The project involves the promotion and installation of improved cook stoves (ICS) in Malawi for use by residential households in rural areas. The ICS employed under this project is the TLC Rocket Stove which substantially reduces fuel consumption and emissions for conducting cooking and water heating tasks in homes. The ICS improve the efficiency of combustion and thermal transfer to the pot compared with a traditional pot support or three-stone fire by incorporating a number of cutting edge components, including one or more of; a 'rocket elbow'; a highly-insulated combustion chamber which provides a conducive environment for clean and efficient combustion of wood. It substantially reduces wood fuel consumption compared with a three-stone fire or traditional pot support.

The project was included as the fifth CPA under CDM PoA entitled Improved Cookstoves Program for Malawi and cross-border regions of Mozambique (CDM PoA reference number -9558, CPA reference number – 9558-0005).¹

The first TLC Rocket Stove was installed on 5 September 2016. Till the end of monitoring period, a total of 4,381 ICSs was reported installed under the project. All the data recorded during ICS registration process was captured via electronic means where the handheld device (PDA) was used. Therefore, no hard copy is available. The information collected via PDA is then transferred to a server which serves as the electronic project database.

CDM CPA inclusion

Date of CPA inclusion into PoA	06-October-2016
--------------------------------	-----------------

Installation of ICS

Date of first ICS installed	05-September-2016
Date of last ICS installed in the database	07-April-2017

First Monitoring Survey

Survey dates for parameters n_y and SS_y	20-April-2017 to 21-April-2017
Survey dates for parameter $\eta_{new,y,i}$	11-April-2017 to 13-April-2017

The ICS disseminated through this project will replace the prevailing inefficient three-stone fires or traditional pot support with stoves which combust wood more efficiently and improve thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions.

The project will take place in Malawi. The details of the project location are provided in Section 1.9.

The conditions existing prior to project initiation are the same as the baseline scenario. See Section 2.4.

The details on Project crediting period are given under Section 1.7.

The project results in a total emission reduction of 6,761 tCO₂e over the monitoring period of 05 September 2016 to 31 July 2017.

¹ <https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

1.2 Sectoral Scope and Project Type

The project is categorised under type/category as below:

a) Sectoral scope: 03 - Energy demand

b) Type: II – Energy efficiency improvement projects

The project is not a grouped project

1.3 Project Proponent

Organization name	Total LandCare (TLC) Malawi
Contact person	Dr. William Trent Bunderson
Title	Co-Founder and Executive Director
Address	PO Box 2440, Lilongwe, Total LandCare, Area 14, Plot 100, Lilongwe, Malawi
Telephone	+265 1 770 904 / 905
Email	total.landcare.mw@gmail.com

Organization name	C-Quest Capital Malaysia Global Stoves Limited
Contact person	Ken Newcombe
Title	Director
Address	Brumby Centre, Lot 42 Jalan Muhibbah, 87000 Labuan, Malaysia
Telephone	+6 087 593828
Email	cqc-operations@cquestcapital.com

1.4 Other Entities Involved in the Project

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

1.5 Project Start Date

05 September 2016 (The earliest ICS installation date under the project).

1.6 Project Crediting Period

05 September 2016 to 04 September 2026 (10 Years , Renewable)

1.7 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	X
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1	44,853
Year 2	44,853
Year 3	44,853
Year 4	44,853
Year 5	44,853
Year 6	44,853
Year 7	44,853
Year 8	44,853
Year 9	44,853
Year 10	44,853
Total estimated ERs	448,530
Total number of crediting years	10
Average annual ERs	44,853

1.8 Description of the Project Activity

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

1.9 Project Location

The project will take place in Malawi. The boundary of the project is determined by the sum of the locations of the individual households within which the ICSs are installed, but all limited to Malawi.



Malawi, Northern Point
Latitude: - 9.366667° S
Longitude: 33.000000° E

Malawi, Western Point
Latitude: - 13.600000° S
Longitude: 32.666667° E

Malawi, Eastern Point
Latitude: -14.883333° S
Longitude: 35.916667° E

Malawi, Southern Point
Latitude: - 17.133333° S
Longitude: 35.283333° E

Map:
http://www.ephotox.com/malawi_region_map.html

Geographical coordinates obtained from Google Earth®

1.10 Conditions Prior to Project Initiation

The conditions prior to project initiation are the same as the baseline scenario. See Section 2.4.

1.11 Compliance with Laws, Statutes and Other Regulatory Frameworks

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

1.12 Ownership and Other Programs

1.12.1 Project Ownership

The letters of Approval (LoA) from Malawi, Netherlands and Sweden have been provided to VVB as the proof of title of the project.

1.12.2 Emissions Trading Programs and Other Binding Limits

Not applicable. The project was registered under UNFCCC.

1.12.3 Other Forms of Environmental Credit

Not applicable.

1.12.4 Participation under Other GHG Programs

The project has been included under CDM POA (Reference number 9558)

1.12.5 Projects Rejected by Other GHG Programs

Not applicable.

1.13 Additional Information Relevant to the Project

Eligibility Criteria

Not applicable. The project is not a grouped project.

Leakage Management

As per AMS-II.G version 5, leakage related to the non-renewable woody biomass saved by the project activity shall be assessed based on ex post surveys of users and the areas from which this woody biomass is sourced (using 90/30 precision for a selection of samples). The potential source of leakage due to the use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources shall be considered. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass by the non-project households/users, that is attributable to the project activity, then B_{old} is adjusted to account for the quantified leakage. Alternatively, B_{old} is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required.

The project opts to apply a net to gross adjustment factor of 0.95 to account for leakages, thus no surveys on leakage are required.

Commercially Sensitive Information

No commercially sensitive information was excluded from the public version.

Sustainable Development

The project contributes to the sustainable development of the Malawian economy in a number of ways:

1. Environmental

- The project will help significantly reduce Malawi's greenhouse gas emissions over its lifetime.

- The project will help reduce the use of non-renewable biomass from Malawian forests, thus assist in conserving existing forest stock and the protection of natural forest ecosystems and wildlife habitats.
- The protection of standing forests will also help protect watersheds that regulate water table levels and prevent flash flooding.

2. Social

- Considerably less time will need to be spent collecting wood fuel for the family home thereby reducing the work burden on rural families and presenting alternative opportunities for economic development
- The amount of indoor pollutants from the burning of biomass in the family home will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to the decrease in total biomass burned and an increase in the temperature of combustion.
- The stove provides a safer method for combusting biomass for cooking, helping to reduce burn injuries, especially for children, in the family home.

3. Economic

- The project will help develop a section of the rural economy in Malawi; in the local assembly, distribution, maintenance and monitoring activities.
- Household expenditures on cooking fuel will be reduced through the use of the ICS
- Saved household labour can be diverted to more productive economic activities.
- Strengthening the employee base of partner organizations and creation of direct local employment opportunities in operational and management roles, as well as future assembly and/or manufacturing initiatives.

The project will deliver a long-term, secure and simple contribution to sustainable development in Malawi that, without carbon finance, would not exist.

Further Information

Not applicable.

2 APPLICATION OF METHODOLOGY

2.1 Title and Reference of Methodology

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

2.2 Applicability of Methodology

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

2.3 Project Boundary

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

2.4 Baseline Scenario

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

2.5 Additionality

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

2.6 Methodology Deviations

Not applicable.

3 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

3.1 Baseline Emissions

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

3.2 Project Emissions

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

3.3 Leakage

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

3.4 Estimated Net GHG Emission Reductions and Removals

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

4 MONITORING

4.1 Data and Parameters Available at Validation

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

4.2 Data and Parameters Monitored

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

4.3 Monitoring Plan

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

5 SAFEGUARDS

5.1 No Net Harm

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

5.2 Environmental Impact

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

5.3 Local Stakeholder Consultation

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

5.4 Public Comments

According to paragraph 3.11.10 of VCS Standard version 3.7, this section does not need to be completed. Please refer to CPA-DD at the website below:

<https://cdm.unfccc.int/UserManagement/FileStorage/O0IWUCZYA1QTERDV7F4LB9MHJX2G5S>

6 ACHIEVED GHG EMISSION REDUCTIONS AND REMOVALS

6.1 Data and Parameters Monitored

Data / Parameter	$n_{y,j}$
Data unit	Quantity
Description	Number of stoves still in operation during the monitoring period as determined by the monitoring survey. This includes total number of stoves installed in the entire project.
Value applied:	4,357
Comments	The percentage of stoves found to be still in operation is applied to the total number of stoves installed in the project (according to the ICS registration records in the monitoring database and the applicable sample frame). The proportion of sampled ICS found to be in operation during each monitoring period will be applied to the total number of stoves in the project when calculating emissions reductions. The percentage of stove found to be still in operation is 99.46%, thus the number of stoves still in operation is determined as below $4,381 \times 99.46\% = 4,357$ units Refer Monitoring survey (sampling) spreadsheet for detailed calculation of the percentage of stove found to be still in operation.

Data / Parameter	$t_{y,j}$
------------------	-----------

Data unit	Fraction
Description	Fraction of monitoring period the stove is in operation (days in operation/total days in monitoring period)
Value applied:	Actual values are unique for each ICS. Average for the project is 0.78
Comments	Refer ER Spreadsheet for detailed calculation of this parameter.

Data / Parameter	$\eta_{new,y,i}$		
Data unit	Fraction		
Description	Continuing efficiency of ICS		
Value applied:	Vintage 1: 0.2728		
Comments	Refer Monitoring survey (sampling) spreadsheet for calculation of the efficiency of TLC Rocket Stove for Vintage 1.		
	The tests were coordinated by the CME and undertaken following WBT protocol 4.2.2 by a trained professional working for the CQC and TLC or an experienced third party. Following equipment's were used.		
	Instrument	Model	Other details
	Weighing Scale	Ohaus Portable Balances – Valor 1000 V11P6-AM	Range: upto 6 kg ($\pm 1g$) Calibration facility: within the instrument with known weights Calibration frequency: once in three years Date of purchase: 21/09/2015 Date of Calibration: 02/02/2017 Validity of calibration till: 02/02/2020
	Digital Thermometer	Fluke 51-2 Single Input Digital Thermometer	Thermocouple Type: Type K, Chromel Alumel, bead style Range: $-40^{\circ}C$ to $+260^{\circ}C$ ($\pm 1.1^{\circ}C$) Calibration frequency: Annual Date of purchase: 21/09/2015 Date of Calibration: 02/02/2017 Validity of calibration till: 02/02/2018

Data / Parameter	SS_y
Data unit	Percentage

Description	The percentage of ongoing baseline stove use within the population of in-use ICS during a monitoring period.
Value applied:	13.04%
Comments	Refer Monitoring survey (sampling) spreadsheet for calculation of percentage of ongoing baseline stove use within the population of in-use ICS during a monitoring period.

6.2 Baseline Emissions

a) Formulae for emission reductions

As per the SSC-POA-DD, emission reductions for the SSC-CPA will be calculated according to the following formula:

$$ER_y = B_{y,savings} \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossilfuel} \times N_{y,i} \times L$$

Where:

ER_y	Emission reductions during the year y in tCO ₂ e
$B_{y,savings}$	Total biomass that is saved in tonnes during the monitoring year(y) per device
$f_{NRB,y}$	Fraction of biomass saved by the project activity in year y that has been established as non renewable biomass (Sub national values for Central, Northern, and Southern Regions are 0.97, 0.93 and 0.90 respectively)
$NCV_{biomass}$	Net calorific value of the non-renewable biomass that is substituted (IPCC default for wood fuel, 0.015 TJ/tonne)
$EF_{projected_fossilfuel}$	Emission factor for the substitution of non-renewable biomass by similar consumers. (IPCC default of 81.6 tCO ₂ /TJ)
$N_{y,i}$	Number of project devices of type i operating in year y
L	A net to gross adjustment factor (0.95 default).is applied above (equation (1) of AMS II.G, version 5) in order to adjust B_{old} to account for leakages as per paragraph 13 (a) of the methodology.

Calculating $B_{y,savings}$

According to the AMS II.G (version 5) methodology, $B_{y,savings}$ may be calculated in a number of ways (as per Options 1, 2 and 3 in Paragraph 12) and this PoA will allow the use of Option 2 in CPAs under this POA.

Option 2.

$$B_{y,savings} = B_{old} \times \left(1 - \frac{\eta_{old}}{\eta_{new,i}} \right)$$

Where:

B_{old}	Baseline Quantity of woody biomass used in the absence of the project activity in tonnes per device
η_{old}	Efficiency of the baseline system/s being replaced. The 0.10 default value is used as the replaced systems are three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves
$\eta_{new,y,i}$	Efficiency of the system being deployed as part of the project activity (fraction)

For the purposes of calculating ex-ante emission reductions a baseline adjustment factor has been applied to B_{old} to account for wood fuel used for the 19.7% of households in the baseline study who reported using a second stove at least once per week. This baseline adjustment factor is based on the mean number of stoves used per household averaged across the entire baseline sample,

calculated to be 1.0471. The value of B_{old} applied in this PoA-DD for wood fuel baseline stoves (8.92 kg/day) incorporates this adjustment factor.

The percentage of households continuing to use a baseline stove in addition to an ICS has been monitored in order to address paragraph 26 (b) of the AMS II.G (version 5) methodology. The monitored (ex-post) percentage of ICS users continuing to use a baseline stove in addition to the ICS (parameter SS_y) was compared to the ex-ante percentage found in the baseline (19.7%) and B_{old} is adjusted proportionally based on the proportional change in the percentage. The parameter used to calculate ex-post $B_{y,savings}$ is $B_{old,adjusted}$ in order to account ex-post for fuel-wood used in baseline stoves in addition to ICS.

If percentage of households continuously using baseline stove is higher than 19.7%, $B_{old,adjusted}$ would become lower than B_{old} ; if the percentage of households continuously using their baseline stove is lower than 19.7%, then $B_{old,adjusted}$ would become higher than B_{old} stated in the PoA DD, in that case B_{old} will be taken into account for calculation of $B_{y,savings}$. If the percentage of households continuously using baseline stoves are found to be the same with the those found during baseline survey i.e. 19.7%, the formula becomes exactly 1. Hence there won't be any change of B_{old} value.

This procedure is outlined here:

In order to account for multiple stoves of different ages (i), which may have different efficiencies ($\eta_{new,y,i}$), this formula can be adapted as follows:

$$B_{y,savings} = B_{old,adjusted} \left(1 - \frac{\eta_{old}}{\eta_{new,y,i}} \right)$$

Where:

$N_{y,i}$ Total number of stoves (i) in operation for a full monitoring period equivalent within each SSC-CPA

η_{old} Efficiency of the baseline system/s being replaced. The 0.10 default value is used as the replaced systems are three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system i.e., traditional stoves.

$\eta_{new,y,i}$ Efficiency of ICS of age i (fraction)

and

$$B_{old,adjusted} = B_{old} \times \left[\frac{1.0471}{1 + (SS_y / 0.197) \times (1.0471 - 1)} \right]$$

Where:

B_{old} Baseline Quantity of woody biomass used in the absence of the project activity in tonnes
 SS_y is the percentage of households that continue to use baseline stoves simultaneously with ICS at least once per week (see section B.7.1. of the SSC-PoA-DD);

Note in the formula above that wood-fuel baseline data is used when calculating $B_{old,adjusted}$. The value 0.197 is the percentage of households in the baseline study who use a second stove simultaneously at least once per week and 1.0471 is the multiple stove adjustment factor, calculated as follows:

$$\text{Multi stove adjustment factor} = \frac{B_{old \text{ adjusted for seasons and multiple stove use}}}{B_{old \text{ adjusted for seasons}}} = \frac{9.34}{8.92} = 1.0471$$

b) Calculation of emission reductions

Calculation of $B_{old,adjusted}$

$$B_{old,adjusted} = B_{old} \times \left[\frac{1.0471}{1 + (SS_y / 0.197) \times (1.0471 - 1)} \right]$$

$$B_{old,adjusted} = 3.2558 \times 1.0471 / [1 + (0.1304/0.197) \times (1.0471 - 1)]$$

$$= 3.3061 \text{ tonnes/stove/year}$$

Calculation of $B_{y,savings}$

$$B_{y,savings} = B_{old,adjusted} \left(1 - \frac{\eta_{old}}{\eta_{new,y,i}} \right)$$

Quantity of woody biomass saved in tonnes per ICS in vintage 1

$$B_{1,savings} = 3.3061 \times [1 - (0.1/0.2728)]$$

$$= 2.0942 \text{ tonnes/stove/year}$$

Emission reductions for each ICS

$$ER_y = B_{y,savings} \times f_{NRBy} \times NCV_{biomass} \times EF_{projected_fossilfuel} \times N_{y,i} \times L$$

The emission reductions shall be considered from the dates of registration of each ICS. CME has opted to present the individual set of emission reductions calculation for each ICS in which the emission reductions obtained under each vintage will be presented separately. The full calculation of each ICS is available in Excel spreadsheet for sharing with DoE.

In this section, only the example of calculation for one of the ICS will be provided. The example given is for ICS with serial number PCI03949 which was registered under CPA 5 (9558-0005).

The details of registration record for ICS PCI03949 is as below.

ICS vintage at end date of monitoring period	Installation date	Serial number	Stove model	User name	Region
1	05/09/2016	PCI03949	TLCRS	ESINATI WILIAM	Southern

Start date of monitoring period for CPA 9558-0005 = 05/09/2016

End date of monitoring period for CPA 9558-0005=31/07/2017

The period of Vintage 1 for ICS PCI03949 is as below (based on Installation Date):

Vintage 1: 05/09/2016 – 04/09/2017

Since, ICS is only eligible to claim VER till end date of monitoring period, thus emission reductions will be claimed for the period stated below.

Vintage 1: 05/09/2016 – 31/07/2017 (330 days or 0.90 year)

Emission reductions achieved under Vintage 1

The CPA has opted for sub national value of f_{NRB} where the value for Southern Region is 0.90. As ICS has operated less than a year of each vintage period, thus fraction of year to be applied in the calculation.

$$\begin{aligned} \text{ERs} &= 2.0942 \times 0.90 \times 0.015 \times 81.6 \times 0.95 \times 0.90 \\ &= \mathbf{1.98 \text{ tCO}_2} \end{aligned}$$

Total emission reductions achieved by the project

To account for the losses of emission reductions (ERs) due to non-operational of ICS, the total of emission reductions achieved by each CPA will take into account the proportion of ICS in operation which 99.46%.

The summary of total net ERs calculation is as below.

Initial total ERs	Calculation of net ERs	Net ERs (rounded down)
6,798.48	6,798.48 x 99.46%	6,761

6.3 Project Emissions

Not applicable.

6.4 Leakage

Not applicable.

6.5 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
Year 2016 (05-Sep-16 to 31-12-2016)	1,344	0	0	1,344
Year 2017 (01-01-	5,417	0	0	5,417

2017 to 31-07-2017)				
Total	6,761	0	0	6,761